Challenge Report

Skin Lesions

Maxime Appert

Introduction

In the realm of medical diagnostics, the accurate classification of skin lesions is paramount for effective treatment planning and patient care. This challenge was about figuring out the best methodology in order to classify skin lesions into eight distinct classes.

This report focuses on my attempts at leveraging image processing techniques, machine learning algorithms and others to achieve robust and accurate classification results.

Convolutional Neural Network

Drawing inspiration from previous coursework, in particular the third part of the MLP TP, I began with data preprocessing and model selection. I of course had to adapt the code to my uses (Creation of a DataLoader so to load the images, modification of the transformations to achieve data augmentation…). I also decided to change the model used in our TP to a pre-trained one, due to its complexity based on many images, it would consistently give better results than any model made in a reasonable timeframe would give me. However, that still left multiple options as torch includes multiple pre-trained models available. Some of the most popular ones include ResNet, MobileNet and EfficientNet. I ended up choosing the Resnet architecture as it is more accurate than any of the other models. This comes at a cost however, Resnet is more computationally expensive than MobileNet and EfficientNet, however, since accuracy was paramount ResNet was preferable. I started on ResNet18, but quickly discovered that ResNet 152 gave significantly better results, and we can confirm this with the chart below.

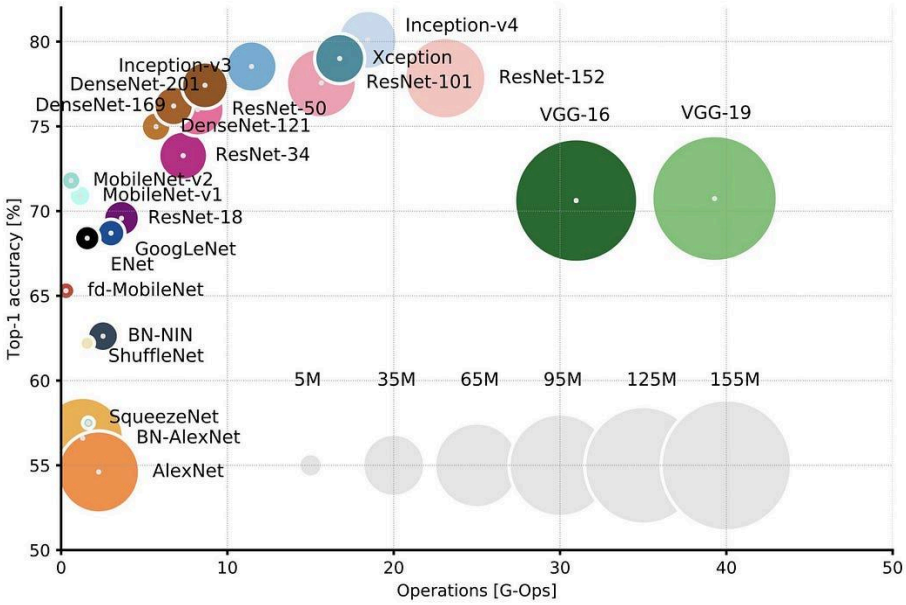


Figure 1: Top-1 one-crop accuracy versus amount of operations required for a single forward pass.

Source: https://towardsdatascience.com/neural-network-architectures-156e5bad51ba

As for the hyperparameters I chose, most of them were figured out through trial and error. This is notably the case for the learning rate, the batch size and the number of epochs. For learning rate, as we have seen in class, we need to choose a balance between too small and too large. Too small, and you will never reach the global minimum while too large can leave you stuck unable to progress due to “bouncing”. To that end, I chose 10^-5, as it gave me the best results I could find. Number of epochs was similarly tweaked in order to avoid overfitting, with 12 epochs being what I found was ideal. Finally, batch size was more elusive, I settled on 32 after testing multiple values, but this could be unoptimized.

My second step in order to improve the accuracy of my model was weighting both the training and testing data appropriately. The weights of the training set were calculated and applied in the cross entropy loss function, while probabilities derived from the testing set's weights were factored into the neural network’s predictions before softmax normalization. This weighting strategy significantly improved classification accuracy.

Afterwards, I tried to incorporate metadata into the classification process, but this proved challenging. My first instinct was to include the metadata directly within the images, but this was very clearly designed for failure as CNNs are adept at extracting features from image pixels but may not inherently understand textual metadata. Additionally, ResNet being a model optimized for classification would not understand this approach. My second instinct was to naively calculate the probability for each class individually for each metadata type, but not only would this not work for age (28, 29 and 30 being represented as independent is not correct), this also means that every metadata type is independent. I then remembered a much better path to use this metadata, I created a decision tree based on the training metadata, and then pondered the softmax made by the neural network with the prediction from the decision tree, which significantly increased the accuracy of my model.

This is the equivalent of bagging, a technique we have seen in class. Indeed, by combining the predictions of multiple weaker models (here 2), we get a model who has a better accuracy. It was interesting to be able to use a technique that I wasn't sure how to implement in this challenge, and it proved to be very useful.

Segmentation

Finally, I tried to implement segmentation, using techniques inspired by our final TP of IMA204, I tried to generate a mask for all the images in which we didn't have access to one. I used opening to try and remove body hairs, I removed the black circles around the images if they were there and after thresholding the images to isolate lesions, I selected the biggest contours. I also noticed that sometimes the contour chosen would simply be the corner of an image, which led me to implement a failsafe in which I created a small square in the center of an image, and if the mask was not present in that square then I would invert the mask. While this did not give ideal results it avoided the cases where nothing could be gained from the image. Despite generally satisfactory results, the neural network would perform more poorly given the segmented images, which is most likely because the segmentation was not consistent enough.

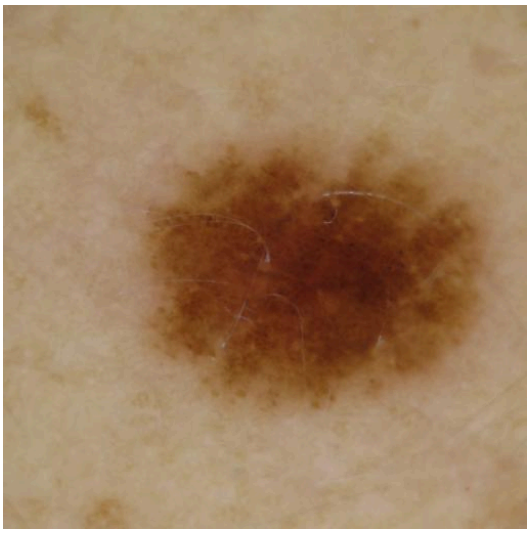


Figure 2: Original Image ISIC_0070948

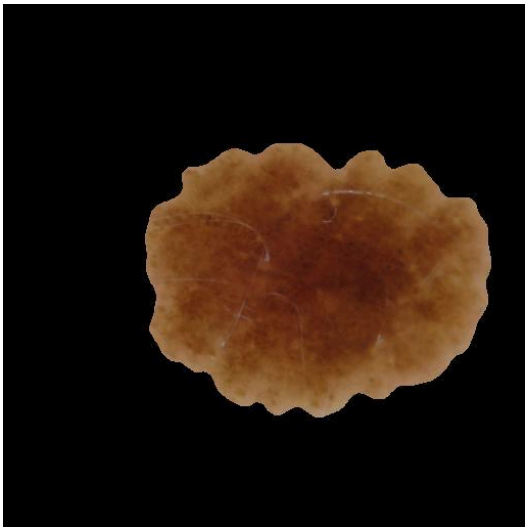


Figure 3: Segmented Image after my Algorithm

Evaluation and Conclusion

My best model's performance was evaluated to around 73%, demonstrating its effectiveness in classifying skin lesions. However, it is clear that some areas could be improved. This is particularly true for segmentation, as we have seen above. One way we could try to improve segmentation is by using SVMs, as they are robust to noise and ideal for finding boundaries and for extracting features. Additionally, exploring more advanced segmentation techniques such as Mask R-CNN could also be beneficial. These methods have shown promising results in medical image segmentation tasks, especially when dealing with complex structures and varying image qualities. Moreover, incorporating ensemble techniques like stacking or boosting could have been interesting in order to improve accuracy and robustness.

Despite these limitations, the approach that I used still scored quite well, and matches my expectations set at the beginning of this challenge from what we had learned in class.